AI for Medical Imaging

Master Artificial Intelligence

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2026/09/15 – Metrics (Part 1)

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Quick announcements

Announcements

Master of AI students: 17:00 info session for the master thesis and thesis fair

Non master-of-AI students: will share separately list of projects from the group

“Horror stories” with labels

Borked annotations and data are unfortunately common

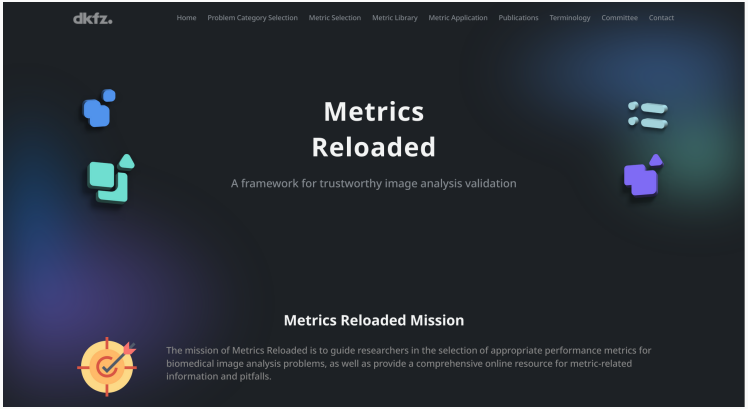
Always inspect and visualise your data

Think if your framing and objective make sense

2026/09/15 – Metrics (Part 1)

Metrics (Part 1)

What metrics to pick?



https://metrics-reloaded.dkfz.de/

Depends on the task...

- Classification: accuracy, sensitivity,
- Localization: IoU, missed boxes, ...
- Segmentation: DSC, Hausdorff distance, ... (today's focus)
- Registration: Guest lecture on October 9

...but also on the specific problem

- multi-class?
- single or multiple connected components?
- *variable* number of connected components?
- shape of the objects?

Overview of the two lectures

Today:

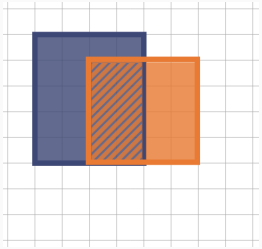
- Dice Score:
  - how to compute its;
  - some of its properties;
  - and its weirdness;
- how to report the metrics;
- and visualize them.

Next week: Hausdorff distance, calibration

2026/09/15 – Metrics (Part 1)

Dice score: deeper dive

Sørensen–Dice coefficient



https://metrics-reloaded.dkfz.de/

$$\mathrm{DSC}(y, s; k) := \frac{2\left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right|}{\left|\Omega_y^{(k)}\right| + \left|\Omega_s^{(k)}\right|}$$

s	discrete segmentation;
$\Omega_s^{(k)} \subseteq \Omega$	subset predicted as k;
$\Omega_y^{(k)} \subseteq \Omega$	subset labeled as k.

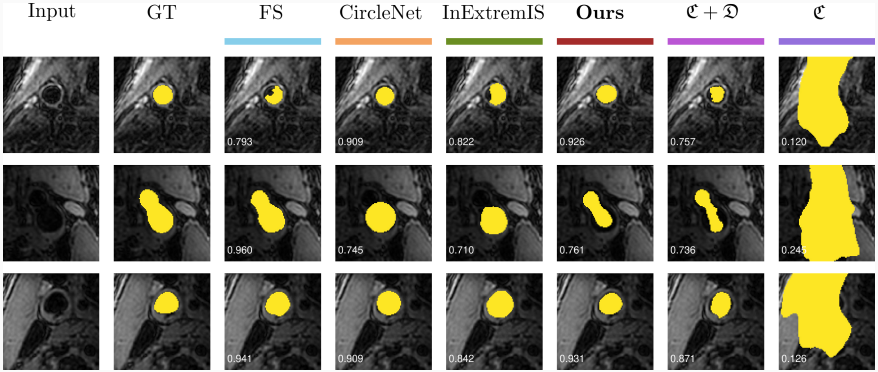
Computing the Dice

$$\mathrm{DSC}(y, s; k) := \frac{2\left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right|}{\left|\Omega_y^{(k)}\right| + \left|\Omega_s^{(k)}\right|} = \frac{2\sum_{i\in\Omega} y^{(i,k)} s^{(i,k)}}{\sum_{i\in\Omega}\left[y^{(i,k)} + s^{(i,k)}\right]}$$

y	discrete ground truth
$y^{(i,k)} \in \{0, 1\}$	
s	discrete segmentation
$s^{(i,k)} \in \{0, 1\}$	
$\mathcal{K}$	set of classes
Ω	image space

https://github.com/HKervadec/ai4mi_project/blob/master/utils.py#L142

Over-segmentation vs under-segmentation



Differentiable Boundary Point Extraction for Weakly Supervised Star-shaped Object Segmentation, *Camaras et al.*, MIDL 2022

Dice favours *over-segmentation* (proof available upon request).

Proof of over-segmentation bias (i)

Let us denote a perfect predicted segmentation S and a ground truth G. The DSC score is henceforth:

$$\text{DSC}(S, G) = \frac{2|S \cap G|}{|S| + |G|} = 1.$$

Let us now denote two errors of the same area (number of voxels) but of different type: $E_1 \not\subset G$ and $E_2 \subset G$, such as $|E_1| = |E_2| = N$.

$S \cup E_1$ represents an *over*-segmentation, whereas $S \setminus E_2$ represents an *under*-segmentation. Notice that both errors, in terms of number of mis-classified pixels, are the same.

Proof of over-segmentation bias (ii)

The corresponding Dice scores are:

$$\begin{aligned}
\mathrm{DSC}(S \cup E_1, G) &= \frac{2|(S \cup E_1) \cap G|}{|S \cup E_1| + |G|} && (6)\\
&= \frac{2|S \cap G|}{|S| + |E_1| + |G|} && (7)\\
&= \frac{2|G|}{2|G| + N} && (8)\\
&= \frac{4|G|^2 - 2|G|N}{4|G|^2 + N^2}, \text{ and} && (9)
\end{aligned}$$

Proof of over-segmentation bias (iii)

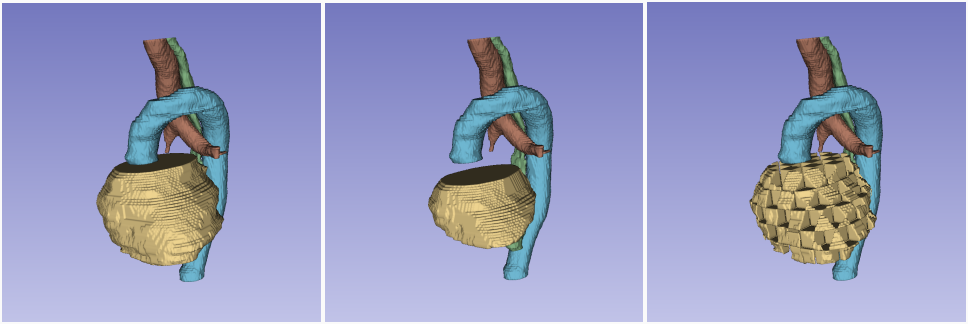
$$\mathrm{DSC}(S \setminus E_2, G) = \frac{2|(S \setminus E_2) \cap G|}{|S \setminus E_2| + |G|} \tag{10}$$

$$= \frac{2(|G| - |E_2|)}{2|G| - N} \tag{11}$$

$$= \frac{4|G|^2 - 2|G|N - 2N^2}{4|G|^2 + N^2}. \tag{12}$$

As both $|G|$ and N are greater than 0, we can therefore show that $\mathrm{DSC}(S \cup E_1, G) > \mathrm{DSC}(S \setminus E_2, G)$; in other words: that for two mis-segmentations of the same area, the over-segmentation will be "favored" by the DSC metric. □

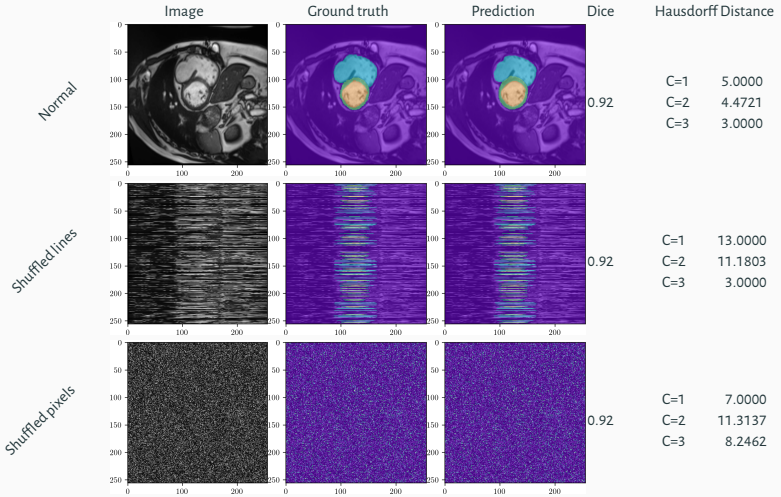
Not all emcompassing



Ground truth | Segmentation 1 | Segmentation 2

Segmentation 1 and 2 have almost the same DSC (≈ 0.66) for the heart.

The loss of spatial relationship



Those pesky corner cases: Dice (i)

	Something to predict	Nothing to predict
I predicted something	Textbook definition	?0
I did not predict anything	?0	?1

Different for all metrics.

Those pesky corner cases: Dice (ii)

Can modify the equation, to avoid an `if/else` in the code:

$$\mathrm{DSC}(y, s; k) := \frac{2\left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right| + \epsilon}{\left|\Omega_y^{(k)}\right| + \left|\Omega_s^{(k)}\right| + \epsilon}$$

Other options (more next week)

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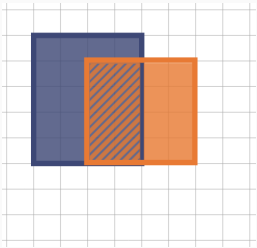
Picking your metrics

Pick several metrics

Always use more than one metric.

But which ones?

Dice and Jaccard (i)



$\Omega_y^{(k)}$, $\Omega_s^{(k)}$

$$\mathrm{DSC}(y, s; k) := \frac{2\left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right|}{\left|\Omega_y^{(k)}\right| + \left|\Omega_s^{(k)}\right|}$$

$$\begin{aligned}\mathrm{Jaccard}(y, s; k) &:= \frac{\left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right|}{\left|\Omega_y^{(k)}\right| + \left|\Omega_s^{(k)}\right| - \left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right|} \\ &= \frac{\left|\Omega_y^{(k)} \cap \Omega_s^{(k)}\right|}{\left|\Omega_y^{(k)} \cup \Omega_s^{(k)}\right|}\end{aligned}$$

Dice and Jaccard (ii)

$$\text{Jaccard}(y, s; k) = \frac{\text{DSC}(y, s; k)}{2 - \text{DSC}(y, s; k)}$$

Method	DSC (average)	Jaccard (average)
M1	93.69	88.47
M2	93.91	88.68

```
>>> 0.9369 / (2 - 0.9369)
0.8812905653278149
>>> 0.9391 / (2 - 0.9391)
0.8851918182675088
```

(Bonus) Proof of Dice and Jaccard relationship

$$\mathcal{D} = \frac{2|A \cap B|}{|A| + |B|}$$

$$\mathcal{J} = \frac{|A \cap B|}{|A \cup B|}$$

$$= \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

$$\frac{1}{\mathcal{D}} = \frac{|X| + |Y|}{2|X \cap Y|}$$

$$\Leftrightarrow \frac{2}{D} - 1 = \frac{|X| + |Y| - |X \cap Y|}{|X \cap Y|}$$

$$= \frac{1}{\mathcal{J}}$$

Pick several metrics *families*

For segmentation:

- overlap based (Dice score and derivatives);
- distance based (Hausdorff distance and derivatives);
- others if relevant:
  - clinical metrics notably.

Browse `https://grand-challenge.org/challenges/` for examples

Metrics: computed with or without post-processing?

If evaluating the method/some new technique, measuring without post-processing might be favourable

If trying to push performances to their limit, with post-processing might make sense

Reporting the two is always an option: can shed insights on the failure case and impact (or not) of post-processing

2026/09/15 – Metrics (Part 1)

Reporting your metrics

At the beginning...

... there was the average.

See https://github.com/HKervadec/ai4mi_project/blob/master/distrib.py

Frameworks for visualization

Matplotlib, notably

https://github.com/rougier/scientific-visualization-book

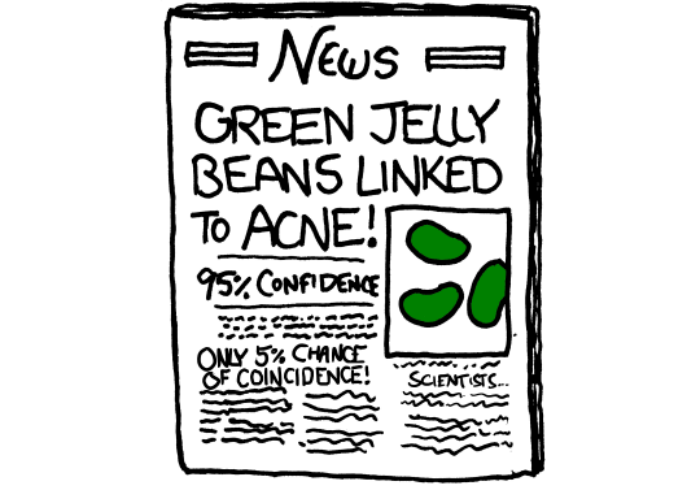
https://seaborn.pydata.org/examples/index.html

https://github.com/plotly/plotly.py

Be creative on how you report and interpret your metrics and their distribution/relationship.

Handling multiple metrics: a complex case study

https://arxiv.org/pdf/2411.08629



https://xkcd.com/882/

Statistical testing

Before doing any statistical testing, pick the definition of the sample carefully (and be clear about it).

Is a sample:

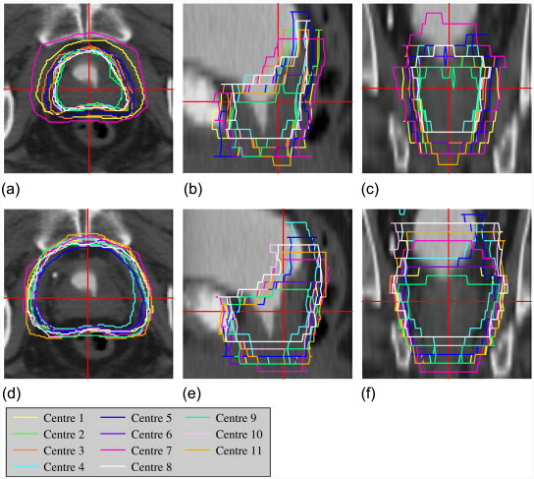
- one run of your K-fold validation (3D DSC averaged over dataset, K of them);
- one scan of a single run (a few dozen depending on dataset);
- one slice of a scan (a few thousands);
- one voxel (a few millions).

The last (two) seriously border on p-hacking.

2026/09/15 – Metrics (Part 1)

The limits of metrics and annotations

The limits of annotations



In many applications, different raters will be north (or even south) of 90% DSC